WEEK 4 · DAY 3 · INSTRUCTOR NOTES

Give the Model Eyes: Feature Engineering

For the trainer only.

The one idea for today: *a model only sees the columns you give it. Give it history (past only), the operating state, and the relationships between sensors, and keep the columns few and sharp.* One good relationship column beats 32 raw ones.

At a glance

Length	~2 hours, including 20 minutes for the final-project Checkpoint 3
Participants leave able to	build lag, difference and rolling features; apply the 14:00 test; build a relationship feature; check a feature on months it never saw; find and prune twin columns; explain a flag with z-scores
Covers (original)	Week 5 Code Book: *Proving EV02 is univariately invisible*, *Proving EV04 needs a derived feature*, *The feature matrix*, *Checking for near-duplicate features*; Week 5 quiz topic *feature engineering from time series* (Quiz 3, questions 16–22)
Datasets	`week-05/data/bus_telemetry_multivariate_hourly.csv`, `week-05/data/anomaly_labels_week5.csv` (both loaded by STEP 0)
Files you need open	animated slides, your **solution** notebook, the rule book PDF, the final-project checkpoint table

The day before: checklist

☐ Run the **solution** notebook top to bottom: 15 of 15 ✅ (under a minute; the forest cells take a few seconds each).

☐ Re-read W4D2's ending: the forest in steady sun noticed EV04 on **1 hour of 672**, 615 hours late. Today starts from there.

☐ Remind participants to bring their final-project method draft (Checkpoint 3), running on **their** data, on NIGCOMSAT systems.

Run of show (2 hours)

Time	Block	What happens	Slides
0:00	**Hook**	NEPA brought light but the inverter isn't charging: EV02 on 2 May	1–4
0:10	**Why features**	Last time: EV04 seen on 1 hour of 672. The model cannot tell time; three kinds of eyes	5–6

Time	Block	What happens	Slides
0:15	**History**	Lags and rolling windows; the 14:00 test; 24 rows lost	7–9
0:22	**Relationships**	Raw temperature sleeps through EV04; `temp_per_drive` lights 605 hours; the 29σ trap	10–13
0:32	**Fewer, better**	18 twin pairs; 32 columns vs 5; right answer, right reason	14–17
0:38	**Cliffhanger**	5 of 5, but 40 false alarms and an exam it marked itself	18
0:40	**Hands-on**	Part A (15 min), Part B (20 min), Part C (20 min)	20
1:35	**Checkpoint 3**	one by one: method drafted (see below)	19
1:55	**Wrap-up**	recap; tease *No expo!* and Quiz 3 next time	21

Explain it like this (in plain English)

- **What is a feature?** Just a column in the table the model reads. Raw sensor readings are features; so is anything you compute from them. Feature engineering = building the columns that let the model see what an engineer already knows.
- **"The model cannot tell time."** The isolation forest reads one row (one hour) at a time and compares it with other rows. It has no idea what happened an hour ago, or whether this is a sunny hour or an eclipse hour, unless a column says so.
- **History features.** `soc.shift(1)` copies the value from one hour earlier onto each row (a *lag*). `soc.diff(1)` is the change since the last hour. `soc.rolling(24).mean()` is the average of the last 24 rows, ending at the current one. All of these look **back**.
- **The 14:00 test.** At 14:00 in the control room, you only have data up to 14:00. `shift(-1)` copies the **next** hour back; `rolling(24, center=True)` uses 12 future hours. Both look brilliant in a notebook and are impossible live. In the exercise, both already "know" about EV02 at midnight, an hour before it began.
- **Rows lost.** A 24-hour window can't be filled for the first 23 hours, and the 24-hour lag needs a full day before it, so `dropna()` removes the first 24 rows: 8,760 → 8,736.
- **Why `temp_per_drive` works.** The amplifier's temperature rises with the drive level (the load) and with the season, so its raw values swing a lot. That swing makes σ big, and the 3σ ceiling (68.0 °C) sits above even the fault (hottest EV04 hour 67.0 °C). Divide by the drive and you ask a sharper question: *how hot is it for the work it's doing?* Normally 0.821 ± 0.013; during EV04, 0.870 (about +3.9σ). A plain 3σ chart on that one column catches EV04 in its first hour, lights **605 of its 672 hours**, and raises **0 false alarms all year**.
- **The 29σ trap.** The code book builds a cleverer feature: an *expected* temperature from drive plus a yearly sin/cos wave, fitted on January–May, and then the residual. EV04 sits **28.9σ** out. Impressive, until you look at the rest of the year: from June it drifts, and from October **every one of the 2,208 hours** is flagged (3,512 false alarms in the year). Why: the amplifier also warms slowly through the year (about +3.7 °C a year in this file), and five months can't tell that slow warming from the seasonal wave. The fitted wave guesses wrong for months it never saw. **The test of a feature is not how far out the fault sits; it's whether the feature stays quiet on normal months it never saw.**

- **Twins.** The code book's 32 columns contain 18 pairs above 0.98 correlation. `soc_gap` is exactly `100 − battery_soc_pct`. The forest picks a column at random for every cut, so two copies of a sensor give that sensor double the vote, and the noise of 32 columns drowns the obvious faults.
- **Lean 5 vs kitchen sink (same forest, same reference months, same steady-sun hours, whole year).** Kitchen sink: 3 of 5 (misses EV01 and EV03, the easy 11 A dropouts), about 70 false alarms. Lean 5 (array current, bus voltage, state of charge, battery temperature, `temp_per_drive`): **5 of 5**, every event within an hour of its start, **40** false alarms.
- **Right reason.** z = (value − normal) ÷ σ, with "normal" from the training months in steady sun. For each event, the biggest z-scores of the flagged hours: EV01/EV03 array current about −34σ and battery temperature about +25 to +27σ; EV02/EV05 state of charge −21.6σ and bus voltage about −5.5 to −5.9σ; EV04 `temp_per_drive` +3.8σ alone. The model's reason is the engineer's reason.
- **Why "steady sun" matters (context, from last time).** Eclipse (battery power) and the 3 recharge hours after it are normal states with their own normal. SoC between 92 and 94% happens on 35 eclipse-night hours a year; in steady sun it happens only during EV02 and EV05.

Slide talking points

#	Say this	Ask this
3	"NEPA has brought light, but the inverter still shows 93% and won't climb." Each number is normal alone.	"Is 93% a problem?" (at night, no; in sunshine, yes)
4	Gold window = EV02, 2 May. The code book's proof looked at 1 May, where SoC is a normal 100%.	–
5	The forest "caught" EV04 on one hour, 25 days late. By the rules it counts; to an engineer it's a miss.	"Would you sign that off?"
6	Quiz Q16. Each row alone. Features carry history, context, relationships.	–
9	Quiz Q19, Q20, Q22. Read the test aloud and get the room to repeat it.	"Which of these could you compute at 14:00?"
11	The ceiling sits above the fault because σ is swollen by the seasons and the load.	–
12	Quiz Q17. Generator running hot on one bulb: the radiator is blocked.	"What else in your unit is only abnormal *for its load*?"
13	The code book says about 22σ; its own recipe gives 28.9σ (Quiz 3 Q18 uses the real number). Either way it's huge, and either way it fails from October.	"Would you put this on the console?"
15	Quiz Q21. Two copies of SoC = SoC votes twice.	–
16	The sink misses the two **easiest** faults. Fewer, sharper eyes.	–
17	Quiz Q14. Point at EV04: one feature, the right one.	"What sentence do you give the duty engineer?"

#	Say this	Ask this
18	Leave them hungry: the score includes the training months, and the simple chart never raced.	"What would make this an honest exam?"

Answer key

Task	Answer	Common mistake → what to say
A1	`HOURS_BACK = 1`	`-1` copies the next hour back: that's the leak in A3
A2	`WINDOW = 24`	`1` (that's just the value itself)
A3	`{"b", "d"}`	Picking `c`: a trailing rolling mean only uses the past, so it's honest
A4	`len(bus) - len(F)` = 24	Guessing 23: the 24-hour lag (`shift(24)`) needs a full day before it
B1	`bus["twta_temp_c"] / bus["rf_drive_pct"]`	Upside down (values near 1.2): temperature goes on top
B2	`K = 3` → caught 1 of 5 (EV04), 0 false alarms	Expecting it to catch the battery events: it watches the amplifier only
B3	`605`	Typing 672 (the event's length, not the hours lit)
B4	`resid_flags.loc["2025-10-01":].sum()` = 2,208 (all of them)	–
B5	`False`	–
C1	`CUTOFF = 0.98` → 18 pairs	–
C2	`CONTAMINATION = 0.005` → 3 of 5 caught, about 70 false alarms	Writing `0.5` flags half of all hours; writing `5` makes sklearn refuse. It's a fraction: `0.005`
C3	`FIT_ROWS = train & steady` → 5 of 5, 40 false alarms, 358 EV04 hours lit	`steady` alone fits on the whole year, EV04 included: EV04 becomes "normal" (only 5 hours lit, 4 of 5 caught). The check rejects it
C4	`"temp_per_drive"`	–
C5	`False`: it trained on January–August, so EV01–EV03 sit inside its training months, and nobody raced the Week 4 chart yet	–
🌶️	`CHART_FIT_ROWS = train & steady` → 5 of 5, **3** false alarms	Keep it quiet: Week 5 Day 1 runs this race honestly on unseen months

Version note. The forest counts (C2, C3) can shift by a few between pandas or scikit-learn versions. We checked pandas 2.2 and 3.1 with scikit-learn 1.6 and 1.9: everything matches except the kitchen sink (69 or 73 false alarms). The checks accept a range. The chart numbers (B2, BONUS) are exact.

🏁 Final project · Checkpoint 3: 🛠️ Method drafted (20 min)

Matches `week-6/final-project/Final-Project-Requirements.pdf`. Go round one by one:

Check	Pass if
Method runs	the **minimum method for their brief** runs end to end on their data: detection (A, E) = a statistical baseline **plus** a multivariate detector on trailing features; degradation / capacity (B, D) = a trend-to-threshold estimate; forecasting (C, E) = a baseline forecast **plus** a regression
Normal in writing	the states or modes of their process (eclipse, rain, weekday/weekend, shift change…) and what they take away first (season, drive level, ageing)
Features	listed, each passing the **14:00 test**; at least one carries history or a relationship
Reference period	months their unit agrees were quiet, written down with the reason
Baseline named	the simple method they will race at the next checkpoint (control chart, threshold, naive forecast)
Data rules	real data on NIGCOMSAT systems only (not Colab); nothing raw on GitHub

A first version that runs beats a perfect plan. Anyone still on synthetic data needs **written** agreement from their unit head. Next checkpoint: **Week 5 · Day 3: honest test** (split by time + a baseline comparison table).

Troubleshooting

Problem	Fix
`NameError: name 'steady_sun' …` / `'score_events'`	They skipped the 🧰 toolbox cell at the start of Part B
`NameError: name 'F' …` in Part C	They skipped the `make_features` cell in Part A
B1 fails with values near 1.2	Division upside down
C2 raises `ValueError` about contamination	They wrote a percentage (`5`). It's a fraction: `0.005` . (`0.5` runs, but flags half the year)
C3 check fails although 5 of 5 caught	They fitted on `steady` (the whole year). EV04 then sits inside the training data and only a few of its hours light up
A neighbour's forest numbers differ slightly	Library versions; fine, the check allows a range

Problem	Fix
Fast finishers	🌶️ race, then: "drop `temp_per_drive` from `X` and refit: what happens to EV04?" (Still 5 of 5, but EV04 lights only 42 of 672 hours, first flag 39 hours late, and false alarms double to 82)

Notes on the original material

- **The EV02 date.** The code book's *Proving EV02 is univariately invisible* checks **1 May** (SoC a normal 100% all day). The label file and the data put EV02 on **2 May 01:00–12:00**: array current 26.93–27.14 A, SoC 92.5–93.4%. The slides and the notebook use 2 May.
- **"Univariately invisible."** True only for array current, the one channel the proof checks. Bus voltage sags to 50.12–50.24 V (normal 50.40 ± 0.04 V), so a sunlit-only 3σ chart on voltage catches EV02 and EV05 with no false alarms. Slide 17 shows the voltage z-scores. Only EV04 truly needs two sensors. (Review file, 12g.)
- **The 22σ claim.** The code book says residual σ ≈ 0.18 °C and EV04 ≈ 22σ. Its own recipe gives σ **0.162 °C** and **28.9σ**, and the same feature flags all 2,208 hours of Oct–Dec. Quiz 3 Q18 is corrected: it gives the real number and asks why the course uses `temp_per_drive` instead. (Review file, 12h.)
- **The 32 columns.** `soc_gap` duplicates `battery_soc_pct` exactly and 18 pairs sit above 0.98, the code book's own pruning line, yet it fits on all of them. On steady-sun hours that forest misses EV01 and EV03 and raises about 70 false alarms. (Review file, 12j.)
- **Why we judge steady-sun hours only.** Eclipse and recharge hours are known operating states; judged as faults, they swamp every detector (W4D2). We state this limit openly: a fault during an eclipse hour is left to the eclipse procedures.